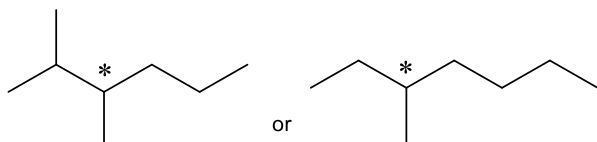
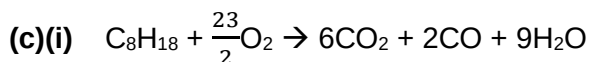
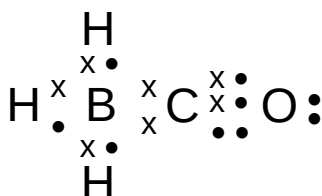


Question 1

- (a) Carbon monoxide bonds strongly (almost irreversibly), via a dative covalent bond, to the iron in haemoglobin. CO is a stronger ligand than O₂ and its presence destroys the O₂ carrying capacity of haemoglobin. Thus, CO is poisonous.

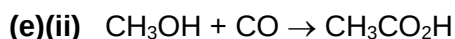
(b)



(c)(ii)

- (d) The presence of ligands causes the splitting of d-orbitals of transition metals into different energy levels. Since the d-orbitals are partially filled, electrons from the lower energy level d-orbitals can absorb energy corresponding to certain wavelengths from the visible spectrum and get promoted to the higher energy d-orbitals. The colour observed is the complement of the colour absorbed.

- (e)(i) Homogeneous catalysis.
The catalyst is in the same phase as the reactants.



- (e)(iii) Steps 3 & 5.

- (e)(iv) **A** is a catalyst as it is consumed in step 1 but regenerated in step 6, showing that it is not chemically changed in the reaction.
B is a catalyst as it is consumed in step 2 but regenerated in step 5 showing that it is not chemically changed in the reaction.
C is a reactant as it is consumed in step 4 but not regenerated at any steps.

(f)(i) $K_p = \frac{P_{\text{CH}_4} P_{\text{H}_2\text{O}}}{P_{\text{CO}} (P_{\text{H}_2})^3}$

Units: atm⁻²

(f)(ii) $P_{\text{CO}} = 0.19 \times 32 = 6.08 \text{ atm}$

$$P_{\text{CH}_4} = 0.12 \times 32 = 3.84 \text{ atm}$$

$$P_{\text{H}_2\text{O}} = P_{\text{CH}_4} = 3.84 \text{ atm}$$

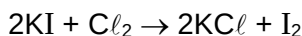
$$P_{\text{H}_2} = 32 - 6.08 - 3.84 - 3.84 = 18.24 \text{ atm}$$

$$(\text{or } P_{\text{H}_2} = 3 P_{\text{CO}} = 3 \times 6.08 = 18.24 \text{ atm})$$

(f)(iii) $K_p = \frac{P_{\text{CH}_4} P_{\text{H}_2\text{O}}}{P_{\text{CO}} (P_{\text{H}_2})^3} = 3.9965 \times 10^{-4} \text{ atm}^{-2}$
 $= 4.00 \times 10^{-4} \text{ atm}^{-2}$

Question 2

- (a) When aq KI is added to aq Cl₂, the colour of the solution changes from colourless to brown due to a redox reaction between KI and Cl₂:



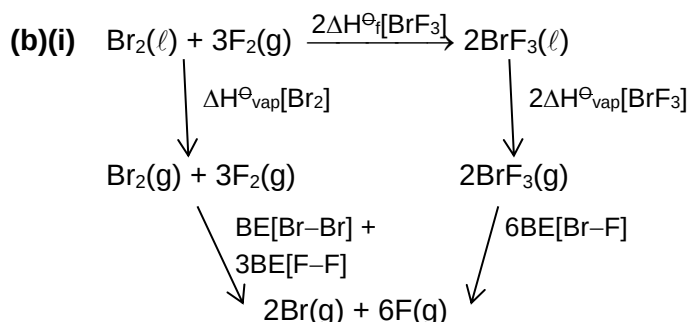
$E^\ominus_{\text{cell}} = E^\ominus_{\text{Cl}_2/\text{Cl}^-} - E^\ominus_{\text{I}_2/\text{I}^-} = 1.36 - 0.54 = +0.82 \text{ V} > 0$
so that the reaction is spontaneous under standard conditions.

When aq KCl is added to aq Br₂, no colour change is observed as Cl⁻ is not able to reduce Br₂ to Br⁻ as predicted by the negative $E^\ominus_{\text{cell}}$:

$$E^\ominus_{\text{cell}} = E^\ominus_{\text{Br}_2/\text{Br}^-} - E^\ominus_{\text{Cl}_2/\text{Cl}^-} = 1.07 - 1.36 = -0.29 \text{ V} < 0$$

When aq KBr is added to aq I₂, no colour change is observed as Br⁻ is not able to reduce I₂ to I⁻ as predicted by the negative $E^\ominus_{\text{cell}}$:

$$E^\ominus_{\text{cell}} = E^\ominus_{\text{I}_2/\text{I}^-} - E^\ominus_{\text{Br}_2/\text{Br}^-} = 0.54 - 1.07 = -0.53 \text{ V} < 0$$



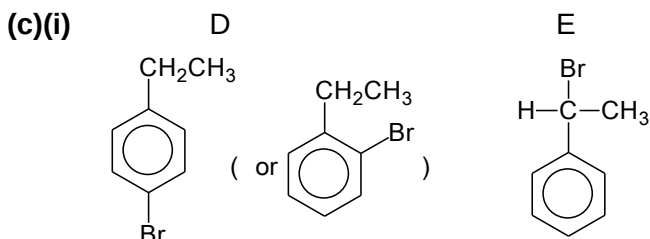
$$2\Delta H^\ominus_f[\text{BrF}_3] = \Delta H^\ominus_{\text{vap}}[\text{Br}_2] + 6\text{BE}[\text{Br}-\text{Br}] + 3\text{BE}[\text{F}-\text{F}] - 6\text{BE}[\text{Br}-\text{F}] - 2\Delta H^\ominus_{\text{vap}}[\text{BrF}_3]$$

$$2(-301) = 31 + 193 + 3(158) - 6\text{BE}[\text{Br}-\text{F}] - 2(44)$$

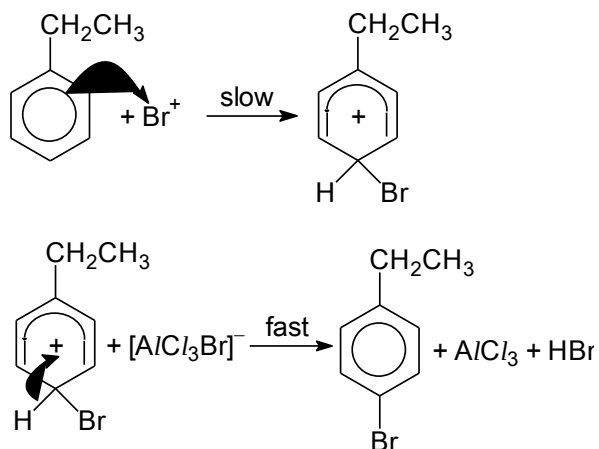
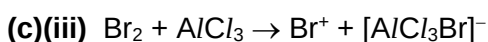
$$6\text{BE}[\text{Br}-\text{F}] = 1212$$

$$\text{BE}[\text{Br}-\text{F}] = 1212/6 = +202 \text{ kJ mol}^{-1}$$

- (b)(ii) $\Delta G^\ominus_f = \Delta H^\ominus_f - T\Delta S^\ominus_f$
 $-241 = -301 - 298(\Delta S^\ominus_f)$
 $\Delta S^\ominus_f = \frac{-301+241}{298} = -0.201 \text{ kJ mol}^{-1} \text{ K}^{-1}$
 $\Delta S^\ominus_f$ is negative which is expected as the reaction results in a decrease in the number of moles of gases, which reduces entropy.

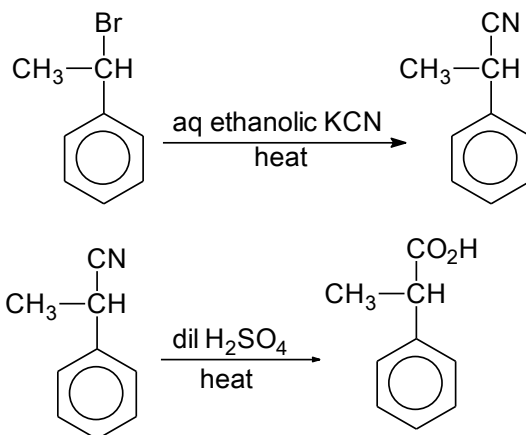


- (c)(ii) Mechanism for reaction 1 is electrophilic substitution while mechanism for reaction 2 is free radical substitution.

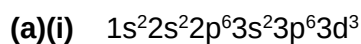


- (c)(iv) Compound E is suitable for the synthesis.
 Step 1:
 Reagent: Aqueous ethanolic KCN
 Conditions: Heat with reflux

Step 2:
 Reagent: Dil H_2SO_4 (or dil HCl)
 Conditions: Heat with reflux

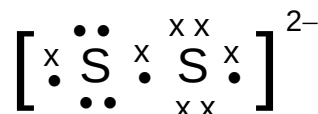


Question 3

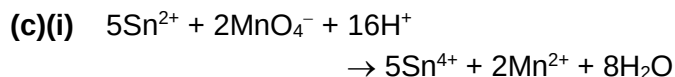


- (a)(ii) MnO_2 . $|LE| \propto \left| \frac{q_+ \times q_-}{r_+ + r_-} \right|$. Since O^{2-} has a smaller anionic radius than S^{2-} , MnO_2 has a more exothermic lattice energy than MnS_2 .

- (b)(i)

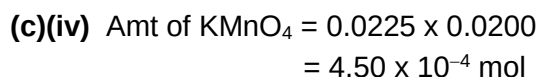


- (b)(ii) Mn^{4+} (Oxidation number +4) undergoes reduction to gain 2 electrons to form Mn^{2+} (oxidation number +2).
 Each S^{2-} (oxidation number -2) undergoes oxidation and loses an electron each, to form S_2^{2-} (oxidation number of each S is -1).



- (c)(ii) $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$ $E^\ominus = +1.52\text{V}$
 $\text{Pb}^{4+} + 2\text{e}^- \rightarrow \text{Pb}^{2+}$ $E^\ominus = +1.69\text{V}$
 $E^\ominus_{\text{cell}} = 1.52 - 1.69 = -0.17 \text{ V} < 0 \text{ V}$
 Hence the reaction is not energetically feasible.

- (c)(iii) Colorless solution turns pale pink.



$$\begin{aligned}\text{Amt of Sn}^{2+} \text{ oxidised} &= 5/2 \times 4.5 \times 10^{-4} \\ &= 1.125 \times 10^{-3} \text{ mol}\end{aligned}$$

$$\begin{aligned}\text{Amt of Sn}^{2+} \text{ in } 250 \text{ cm}^3 &= 1.125 \times 10^{-3} \times 10 \\ &= 0.01125 \text{ mol}\end{aligned}$$

$$\begin{aligned}\text{Mass of Sn}^{2+} \text{ in sample} &= 0.01125 \times 118.7 \\ &= 1.335 \text{ g}\end{aligned}$$

$$\begin{aligned}\% \text{ by mass in sample} &= 1.335 / 3.00 \times 100\% \\ &= 44.5\%\end{aligned}$$

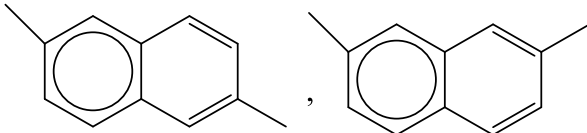
(d)(i) Compound **F** contains a tertiary alcohol group which cannot be oxidised by acidified $\text{K}_2\text{Cr}_2\text{O}_7$ but will undergo acid-metal displacement with a reactive metal like Na. **F** cannot be a carboxylic acid as it contains only one O and **F** cannot be a phenol as it does not have a benzene ring.

(d)(ii)

F		G	
H		I	
J		K	
L			

(d)(iii) enantiomerism and cis-trans isomerism
 $2^2 = 4$ stereoisomers

(d)(iv)

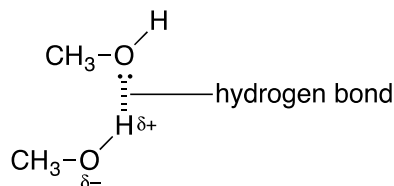


(Other structures are also possible.)
 positional isomerism

Question 4

(a) Ethane and methane are both non-polar hydrocarbons with instantaneous dipole-induced dipole (id-id) interactions between their molecules. Electrons are constantly moving and at any given moment, the electron density of a molecule can be unsymmetrical, resulting in an instantaneous dipole, which induces a short-lived dipole in a neighbouring molecule. Id-id interactions become stronger when the electron cloud increases in size and becomes more easily distorted. Ethane is a larger molecule with more electrons and thus a larger electron cloud size. As a result, ethane experiences stronger id-id interactions which require a higher energy to overcome and therefore its boiling point is higher.

A methanol molecule contains an $-\text{OH}$ group with H covalently bonded to O. Since O is highly electronegative, it attracts the bonding electrons that it shares with H strongly to itself so that H is partially positive. Since H does not have inner shell electrons, its proton is exposed and interacts strongly with a lone-pair of electron on O in a nearby methanol molecule. The interaction between the partially positive H atom attached to O and lone pair of electrons on O of another methanol molecule gives rise to H bond as shown below:



Since the M_r of ethane and methanol do not differ much, the number of electrons of ethane and methanol are similar, so that they have similar id-id interactions. However, relatively strong H bonding exist between methanol molecules but are absent in ethane molecules. Thus, more energy is required to overcome the

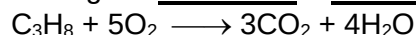
intermolecular forces of attraction in methanol and its boiling point is higher.

4(b)(i) Alkanes are unreactive because

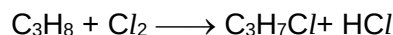
① Alkanes are **saturated**. They do not contain any region of high electron density and thus do not attract electrophilic reagents. They also do not contain any electron deficient sites to attract nucleophilic reagents as C is only slightly more electronegative than H, so the C–H bond is essentially non-polar.

② The strong C–C and C–H bonds in alkanes further contribute to the lack of reactivity of alkanes.

4(b)(ii) When **heated** with **oxygen**, propane undergoes **oxidation** or **combustion**:

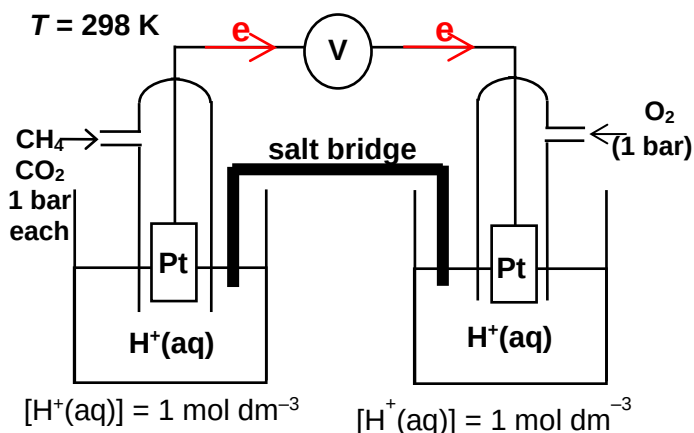


In the presence of **uv light**, propane undergoes **free radical substitution** with chlorine:



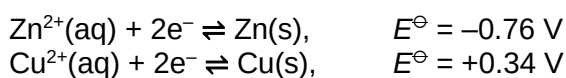
(c)(i) $E^\ominus_{\text{cell}} = 1.23 - 0.17 = +1.06 \text{ V}$

(c)(ii)

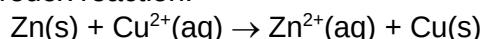


d)(i)

Use $E^\ominus_{\text{cell}}$ values from the Data Booklet,



Overall redox reaction:



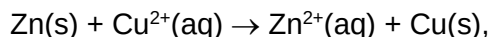
For each pair of zinc and copper discs,

$$E^\ominus_{\text{cell}} = +0.34 - (-0.76) = +1.10 \text{ V}$$

As the voltage pile contains 5 pairs of zinc and copper discs connected in series, total voltage = $5 \times E^\ominus_{\text{cell}} = 5.50 \text{ V}$

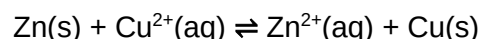
(d)(ii) The mass of the copper discs will have increased while the mass of zinc discs will have decreased.

From the overall redox reaction:



it can be seen that zinc is a reactant and as the reaction progresses, zinc is used up resulting in a decrease in mass of zinc discs. Copper is produced in the reaction and results in an increase in the mass of the copper discs.

The voltage decreases as the concentration of Zn^{2+} increases while the concentration of Cu^{2+} decreases as the reaction progresses:

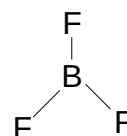


By Le Chatelier's Principle, the position of equilibrium of the above reaction will shift to **left** and hence $E < +5.50 \text{ V}$ (i.e. $E_{\text{cell}} < E_{\text{cell}}^\ominus$).

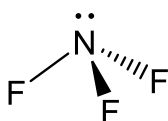
Question 5

(a)(i) VSEPR states that electron pairs (bond pairs and lone pairs) in the **valence/ outer** shell of the central atom are arranged as far apart as possible in space, to minimise their mutual repulsion. Lone pair-lone pair repulsion > lone pair-bond pair repulsion > bond pair-bond pair repulsion.

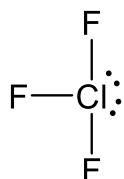
(a)(ii) BF_3 has 3 bond pairs and no lone pairs around B. Hence its electron pair geometry and molecular shape are both **trigonal planar**.



NF_3 has 3 bond pairs and 1 lone pair around N. Hence its electron pair geometry is tetrahedral and its molecular shape is **trigonal pyramidal**.



ClF_3 has 3 bond pairs and 2 lone pairs around Cl . Hence its electron pair geometry is trigonal bipyramidal and its molecular shape is T-shaped.



(b)(i) React the alcohol with either PCl_5 or SOCl_2 at room temperature to convert the alcohol to its corresponding chloroalkane.

(b)(ii) *Reactivity (Ease of hydrolysis):*
ethanoyl chloride > chloroethane > chlorobenzene

Ethanoyl chloride reacts the most readily due to the **higher δ^+ charge on C** (bonded to 2 strongly electronegative atoms, O and Cl), hence attracting nucleophile more strongly. The **highly polarised C–Cl bond cleaves easily** without heating when ethanoyl chloride is reacted with water.

Chloroethane has a lower **δ^+ charge on C** (bonded to 1 electronegative Cl atom), hence attracts nucleophile less strongly than ethanoyl chloride. The **C–Cl bond cleaves only with heating** when reacted with aq NaOH.

The p-orbital on Cl atom of chlorobenzene overlaps with π -electron cloud of the benzene ring so that the **C–Cl bond has partial double bond** character. Thus, no cleavage occurs due to strengthening of the C–Cl bond and chlorobenzene does not react even when heated with aq NaOH.

(c)(i) Phenol can react with either aqueous chlorine or chlorine in CCl_4 at room temperature.

(c)(ii) The acidity of phenols depends on the stability of the phenoxide ion PhO^- as shown in the equation:



For the chlorinated phenol, as chlorine is an electron withdrawing group, it enhances the delocalisation of the negative charge on O in PhO^- into the benzene ring. Due to better charge dispersal, the chlorinated phenoxide ion is more stabilised than the phenoxide ion, causing chlorophenol to dissociate to a greater extent so that chlorophenol is a stronger acid.

(d)

Step 1

Reagent: aqueous KMnO_4 and dil H_2SO_4

Condition: heat with reflux

Step 2

Reagent: PCl_3 , PCl_5 or SOCl_2

Condition: anhydrous

Step 3

Reagent: HCl

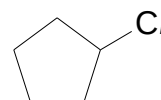
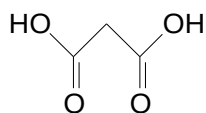
Condition: anhydrous

Step 4

Reagent: Conc NH_3

Condition: in ethanol, heat in a sealed tube

M: N:



P:

